



Flagged Before the Portal Notices

Forecasting Which Warranty Claims Bounce Back for More Paperwork

Fenna Okoro · Ronja Oyelaran — School of Emergent Priorities

The problem

A home-appliance warranty portal rejects a documentation package as “incomplete” days after submission, sending the claim back to the queue it just left. Could the customer have learned this at the moment of submission instead?

Aims

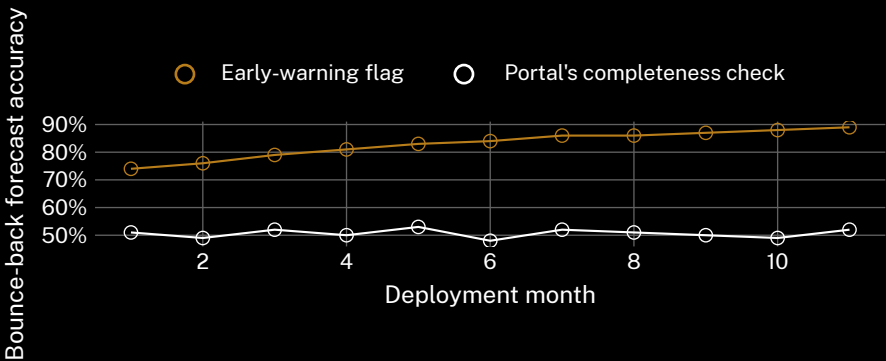
- Build a submission-time signal flag that forecasts documentation bounce-back before the portal’s own automated check does.
- Compare the flag’s forecast accuracy against the portal’s completeness check across eleven months of deployment.
- Track what flagged customers do differently when they resubmit.

Study design

- n = 1,860 warranty claims · 400 households · one national appliance-warranty programme · 11 months
- logistic flag on submission-time signals: photo count, time of day, appliance age band, claim channel
- flag surfaced pre-submission to a randomly held-out subsample; portal’s checklist ran unchanged for all

Findings

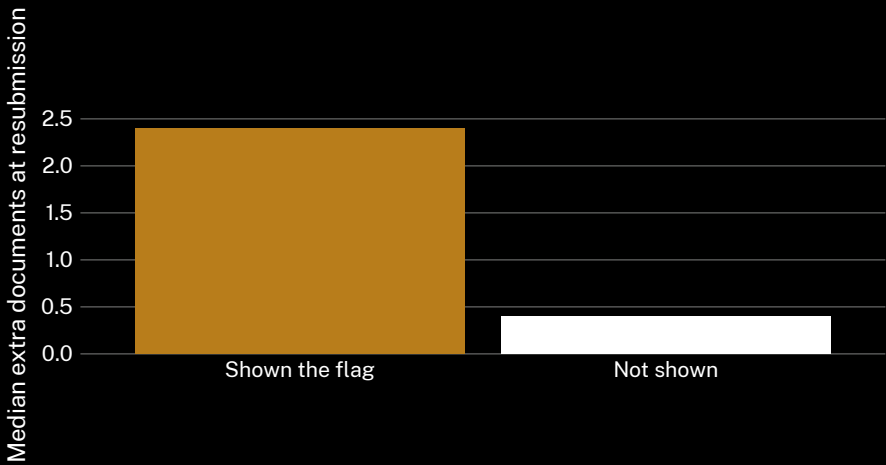
The flag forecasts bounce-back well ahead of the portal’s own completeness check, which tracks close to chance throughout. Customers shown the flag do not bounce back less often; they attach more.



Forecast accuracy for documentation bounce-back rose from 74% to 89% across eleven months of deployment (Jan–Nov 2026); the portal’s own completeness check held at 48–53% throughout.

Interpretation

The flag catches what the checklist counts, not what the reviewing agent actually objects to. Warned customers over-comply rather than correct: a median of 2.4 extra, unrequested documents per resubmission, with no matching fall in bounce-back rate. The portal’s completeness field appears to measure attachment count, not the mismatch an agent later cites. The flag was never asked to reduce paperwork, only to predict it, and the programme’s own success metric — fewer bounce-backs — moved for neither arm. Next: surface the flag’s specific reason, not a generic completeness warning, and test whether that changes what customers attach rather than how much.



Customers shown the flag attached a median of 2.4 unrequested extra documents at resubmission, against 0.4 for customers who weren’t shown it (n = 1,860 claims, 11 months).

Selected references

1. Moynihan, D., Herd, P., & Harvey, H. (2015). Administrative burden: Learning, psychological, and compliance costs in citizen-state interactions. *Journal of Public Administration Research and Theory*, 25(1), 43–69. doi:10.1093/jopart/muu009
2. Gordon, E., Harlow, J., Whitman, S. A., & Lee, M. (2024). Data discretion: Screen-level bureaucrats and municipal decision-making. *Digital Government: Research and Practice*, 5(2). doi:10.1145/3652950
3. Raza, S., Gerke, S., Silcox, C., Hendricks-Sturup, R., & Shachar, C. (2026). Medicare advantage becoming a disadvantage with use of artificial intelligence in prior authorization review. *npj Digital Medicine*, 9, 208. doi:10.1038/s41746-026-02387-x
4. Baran, S., & Rola, P. (2022). Prediction of motor insurance claims occurrence as an imbalanced machine learning problem. arXiv:2204.06109.
5. Beng, C., Hanke, M., & Nwosu, J. (2026). Notified, Not Yet Loaded: Predictive Ready-for-Pickup Alerts Against a Parcel Locker’s Own Door-Sensor Confirmation. *Slo University technical report*. doi:10.5555/slop.yqm3vd

De-identified claim metadata only; no appliance serial numbers or images retained beyond the review window.